PULSE CHECK: UNCOVERING HEART ATTACK RISK FACTORS

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Introduction and Data Overview

Every 40 seconds, someone in the US experiences a heart attack. Over 800,000 Americans have a heart attack every year and it has remained the leading cause of death in the US since the 1950s. Heart attacks occur when blood flow to the heart is blocked, preventing oxygen from reaching the heart muscles and leading to tissue damage or death. And, in the last five years, the rate of heart attacks in the country have been rising, alarming many healthcare professionals and researchers. Studies have been conducted to track patterns in heart attack occurrences and identify preventative measures.

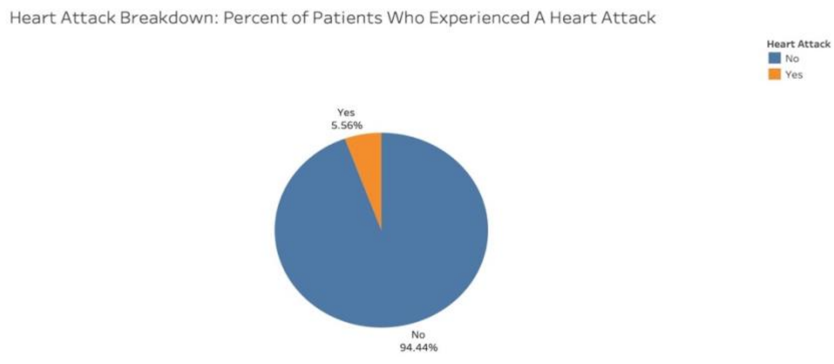
This report aims to analyze the relationship between different factors that impact a person's likelihood of experiencing a heart attack. Specifically, we first look at how gender and geographic location influence the likelihood and distribution of heart attack occurrences across different age groups. Then, we examine if lifestyle choices, such as drinking and smoking, play a role in patients' likelihood of experiencing a heart attack. And finally, the study observes whether a history of clinical conditions, like kidney disease and diabetes, contribute to heart attack. By exploring these questions through graphical visualizations and statistical testing, we hope to identify characteristics that can contribute to increased risk of heart attack, which can be used towards further studies and implementing preventative measures.

For our analysis, we chose a dataset containing medical records for almost 238,000 patients from 2018. With a total of 35 different variables, it offered insights into patient demographics, their clinical conditions, and medical history. This included variables such as age, gender, whether the patient had a heart attack, whether the patient has diabetes, smoking status, and their history of vaccinations. This dataset is ideal because it contains a mix of qualitative, quantitative, and geospatial variables with little to no missing values. For analysis based on location, this dataset was joined with another dataset containing 2018 us state populations on their respective state columns. Additionally, the states values were aliased with state abbreviations for easier readability. No further data transformation or cleaning was needed. This structure allows for exploratory data analysis as well as statistical testing to uncover patterns that can inform healthcare decisions and policies in heart attack occurrences.

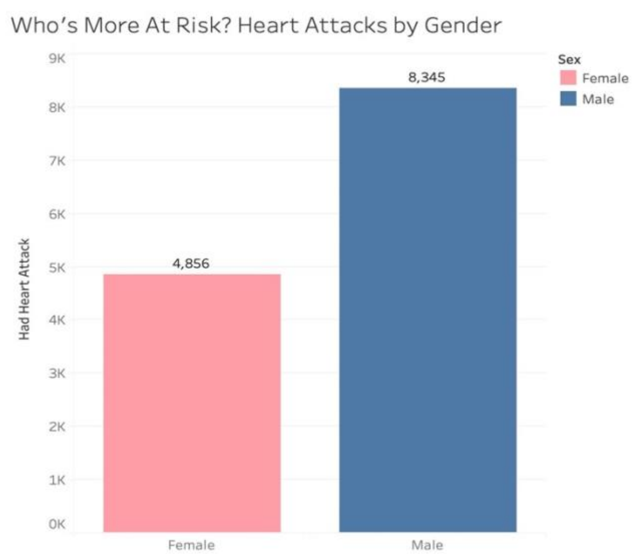
Visualizations and Analysis

For our preliminary exploration of the dataset, we created three charts that provided key insights into heart attacks. First, we found that only 5.56% of the patients experienced a heart attack, highlighting a relatively low occurrence in the dataset. The next chart examined the occurrence of heart attack by gender, demonstrating that men are more affected by women, with 8,345 and 4,856 occurrences respectively. Our final graph analyzed heart attack occurrences by weight, identifying that individuals within the 60-80 kg weight range are at the highest risk.

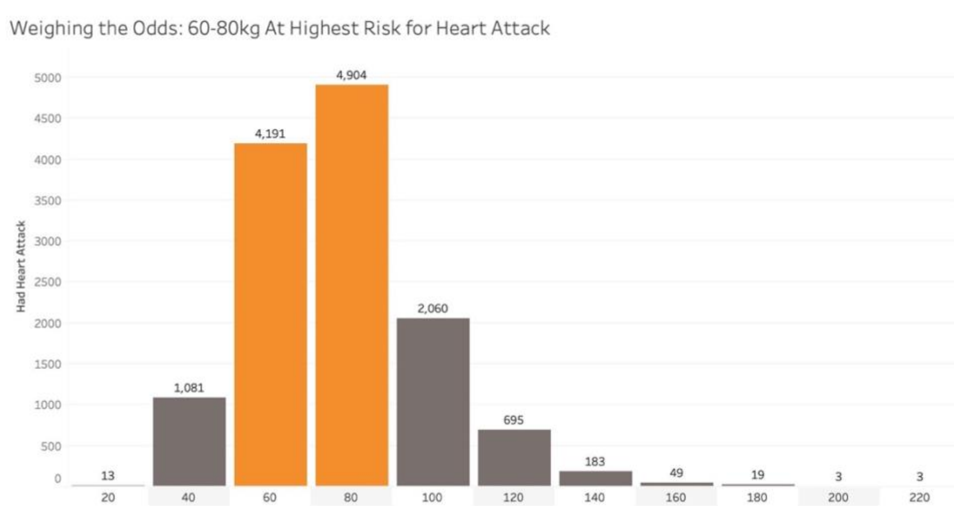
These visualizations provided an insightful overview of trends in heart attack occurrences before we moved into analyzing our research question.



Pie chart showing percentage of patients who experienced a heart attack.

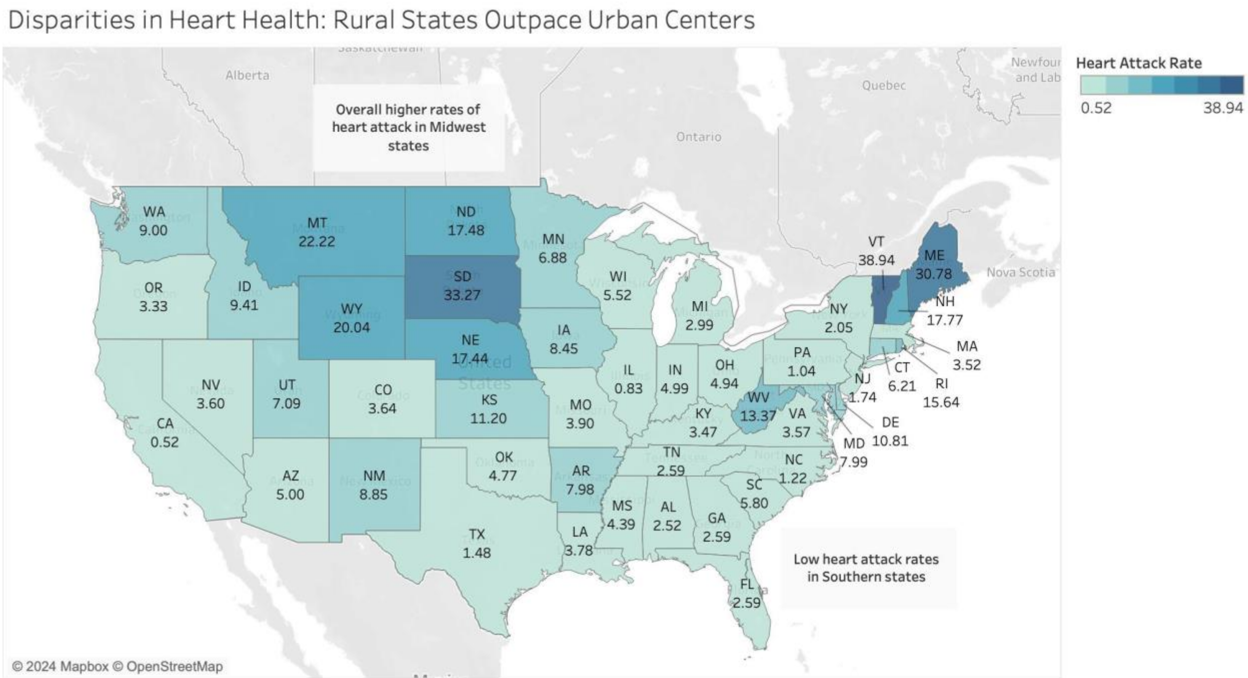


Bar chart showing heart attack occurences by gender.



Bar chart showing heart attack occurence by weight in kilograms.

As part of our first research question to observe how demographic and geospatial factors impact heart attack occurrences, we mapped the rate of heart attack in each US state, per 100,000 people. The map below demonstrates our findings.

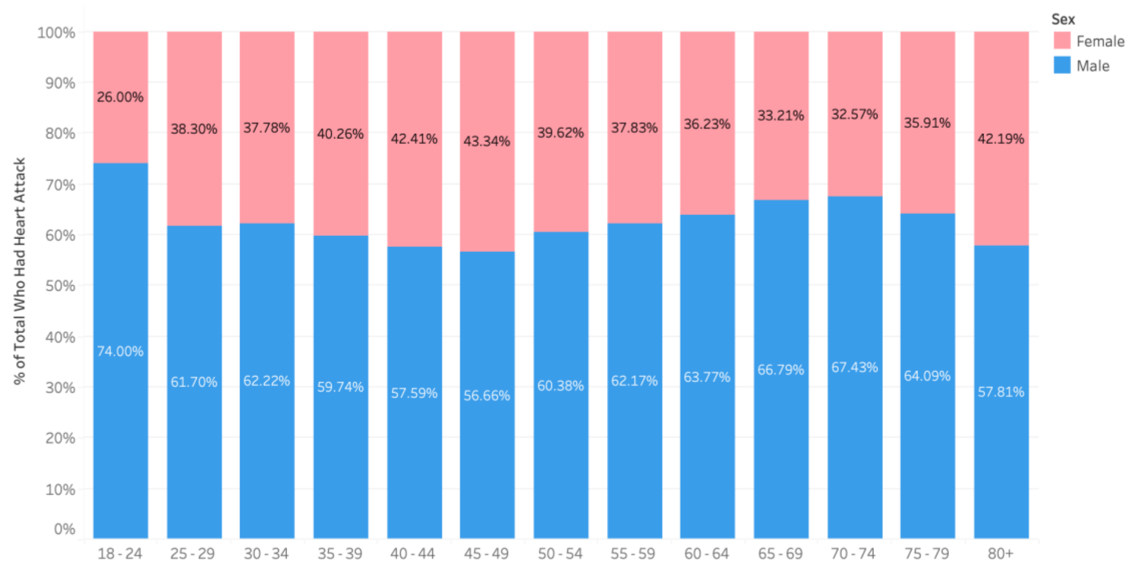


Map depicting heart attack rates per 100,000 people across US states for 2018.

The graph indicates a pattern in heart attack rates based on geographic location. Rural states seem to have higher rates of heart attack, specifically those in the Midwest region of the country and Northeast states like Vermont, New Hampshire, and Maine. In contrast, states with more urbanization, such as Western and Southern states, along with states lower in the Northeast have significantly lower rates of heart attack. A likely explanation for this difference may be the accessibility to healthcare facilities. Rural regions typically have limited access to these facilities, contributing to higher rates. States like California and New York have better healthcare facilities and preventative care, reducing their rate. In addition to access to healthcare, variations in diet and exercise may also play a role in heart attack occurrences. For example, states in the Midwest tend to lean towards more processed and high-fat foods which can decrease heart health. They also have fewer facilities for physical activity and little pedestrian-friendly areas, further declining health. Urban states emphasize the importance of plant-based diets and have better access to healthy grocery store options, along with gyms and parks that promote an active lifestyle.

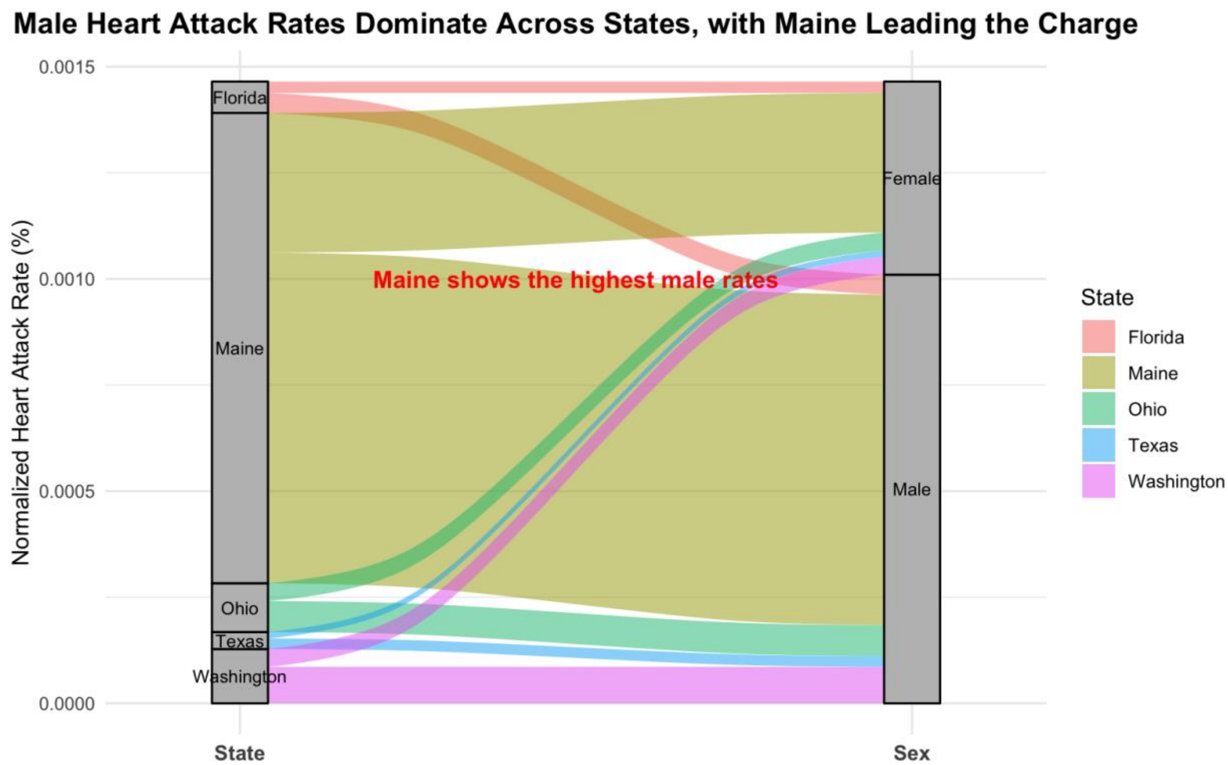
After identifying regional patterns, we next examined demographic factors like gender and age to uncover individual-level disparities in heart attack occurrences.

Men Consistently at Higher Risk of Heart Attack Compared to Women



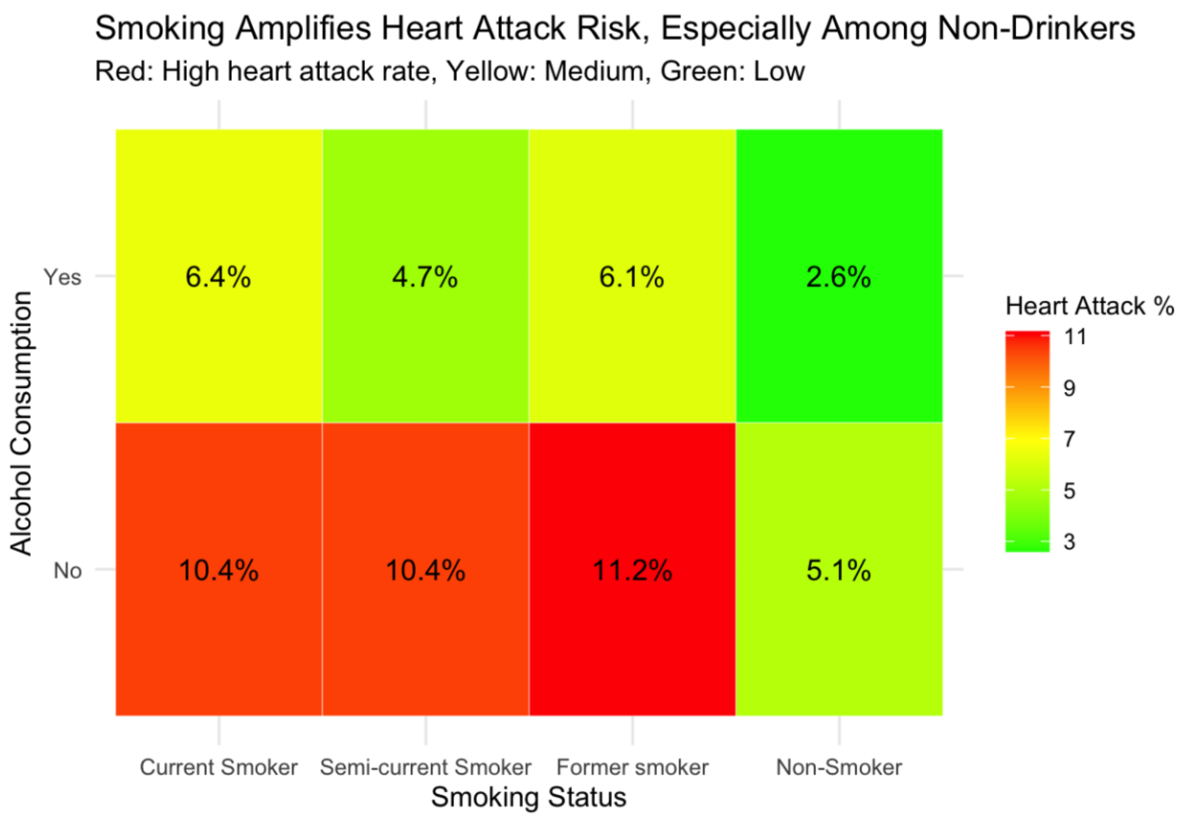
Bar chart showing percentage of men and women experiencing heart attacks in each age group.

Across all ages, the chart shows a higher likelihood of heart attacks for men compared to women. This disparity is particularly apparent between the ages of 18 and 25, where men account for almost three-quarters of the heart attacks. The risk is also high for men between the ages of 65 and 75. This ratio decreases in later adult years, between the ages of 40 and 50. For individuals over the age of 80, the percentage of women experiencing rises to 42.19%, compared to 57.81% for men. But why might this gender and age disparity be the case? For young men, it may be the result of their increased likelihood to engage in high-risk behaviors such as smoking and poor diet. For women, the rise in rates in later adult years may be due to menopause, where decreasing rates of estrogen make them more prone to heart attacks. Furthermore, the risk can increase with stress-related factors, which older women tend to experience.



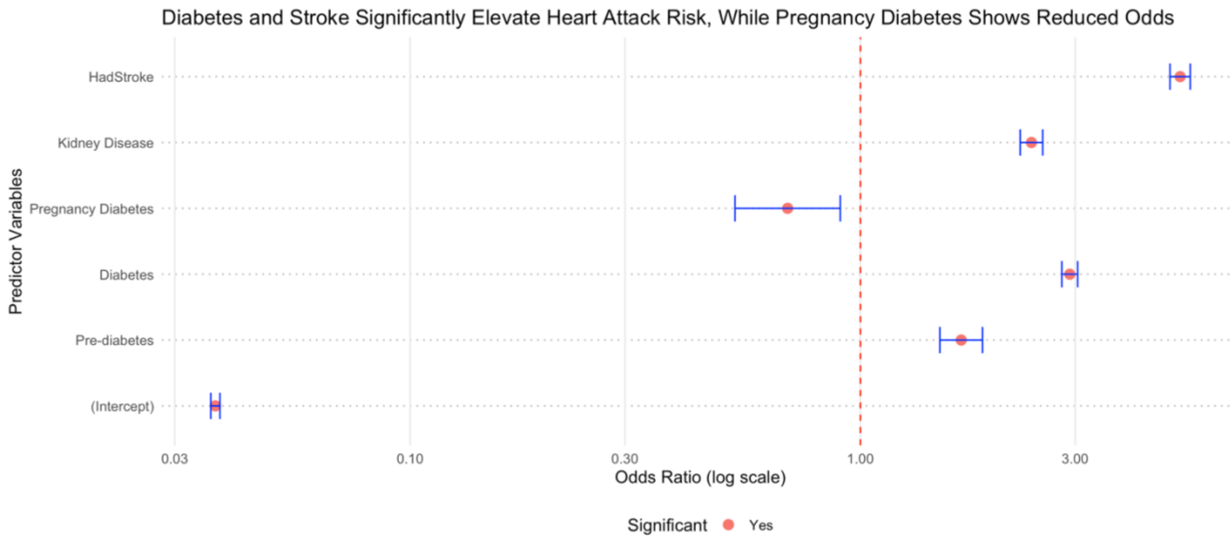
The alluvial graph above provides a clear visualization of gender and state-specific disparities in heart rates. Across all states, men exhibit higher normalized rates of heart attacks than women. Maine emerges as a standout, with the highest rates among men, suggesting possible regional factors such as lifestyle choices, healthcare accessibility, or demographics contributing to this trend. On the other end, Washington shows the lowest heart attack rates for both men and women, hinting at more favorable health or environmental conditions in the region. This gender disparity is evident in every state. But why is this the case? Men are often more prone to engage in high-risk behaviors, such as smoking, heavy alcohol consumption, and unhealthy diets, which contribute to a greater likelihood of heart attacks. On the other hand, women benefit from the cardioprotective effects of estrogen earlier in life, which diminishes after menopause. These differences, combined with potential cultural and socioeconomic factors across states, create a complex picture of cardiovascular health disparities.

Building on these regional trends, we examined individual factors like smoking and alcohol consumption to better understand lifestyle-related risks. Our second research area looked at whether an individual's alcohol consumption and smoking habits could influence their likelihood of experiencing a heart attack, using a heatmap.



We discovered that the highest rates of heart attack were among those who smoked, whether they had formerly smoked, currently smoke occasionally, or smoke regularly. Non-smokers reduced their chances of heart attack by about 50%, regardless of alcohol consumption. Interestingly, this chart indicated that individuals who drank alcohol were less likely to experience a heart attack. Upon reviewing some pre-existing research, the explanation we found is that moderate levels of alcohol consumption do reduce a person's chances of experiencing a heart attack. This is because alcohol can increase the amount of good cholesterol in the body, which helps combat against artery damage. However, high alcohol consumption would significantly increase risk. Our data does not specify how much a patient drinks alcohol, simply whether they do or do not.

Our final question examined if having a history of other clinical conditions, such as kidney disease or diabetes or having had a stroke before, contribute to a heart attack.



This logistic regression model assesses the impact of clinical conditions, including stroke, diabetes, and kidney disease, on the likelihood of a heart attack. Results are adjusted for binary predictors, with odds ratios displayed alongside 95% confidence intervals.

The coefficient plot underscores the significant impact of specific clinical conditions on heart attack risk. Stroke stands out as the most influential factor, with individuals who have experienced a stroke being over five times more likely to suffer a heart attack. Kidney disease and diabetes also pose substantial risks, with odds ratios of 2.40 and 2.92, respectively. Interestingly, gestational diabetes during pregnancy appears to reduce the odds of a heart attack, potentially reflecting temporary hormonal effects or enhanced health monitoring during pregnancy. The differing impacts of these conditions warrant closer examination. Stroke's influence likely stems from the severe damage it causes to the cardiovascular system, while kidney disease reflects long-term systemic health deterioration that heightens vulnerability. Diabetes, particularly unmanaged cases, exacerbates risk through complications such as arterial damage. These findings emphasize the importance of targeted interventions and management strategies to mitigate the cardiovascular risks associated with these conditions.

To evaluate these associations, we performed a logistic regression analysis that assessed the impact of clinical conditions—including diabetes, kidney disease, and stroke—on heart attack risk. The statistical test revealed that stroke, kidney disease, and general diabetes were significant predictors, with p-values well below the 0.05 threshold. The odds ratios further quantified the effects, confirming stroke as the most impactful factor, followed by diabetes and kidney disease. Gestational diabetes, on the other hand, showed a protective association, with odds ratios below 1. This rigorous approach enabled us to draw meaningful conclusions about the relationships between these conditions and heart attack risk.

In constructing this model, we intentionally focused on clinical conditions rather than incorporating additional predictors such as demographic factors (e.g., age or gender). Our goal was to isolate and evaluate the direct cardiovascular risks posed by these conditions without introducing potential multicollinearity or diluting the specific relationships under investigation. However, future models could benefit from exploring demographic and lifestyle factors to provide a more comprehensive understanding of heart attack risk. By narrowing our focus, we were able to identify critical clinical predictors and offer actionable insights into targeted prevention and management strategies.

Findings

Our analysis reveals significant insights into the factors influencing heart attack risks, addressing our research questions through the presented graphs. The geospatial map highlights those rural states, particularly in the Midwest and Northeast (e.g., South Dakota and Vermont), exhibit higher heart attack rates compared to urbanized areas like California and Texas. This disparity likely stems from differences in healthcare accessibility, lifestyle habits, and preventative care availability. Urban areas tend to have better infrastructure for active lifestyles, healthier food options, and robust healthcare systems. These findings suggest that targeted healthcare initiatives are critical in rural areas to mitigate cardiovascular risks. Moreover, we saw in the bar chart analyzing heart attack occurrences by gender and age group shows that men consistently experience higher rates across all age groups, with younger men (18–25) at particularly high risk. This could be attributed to riskier behaviors, such as smoking and unhealthy diets, which are common among young men. On the other hand, the rise in rates among older women post-menopause may be due to reduced estrogen levels and increased stress-related health issues. These findings answer our research question on demographic impacts and stress the importance of gender-specific preventative measures. Lastly, the alluvial chart reaffirms gender disparities in heart attack rates, with states like Maine exhibiting the highest rates among men, potentially due to lifestyle and regional factors. In contrast, Washington's low rates for both genders suggest the influence of effective healthcare policies and healthier living conditions.

We also saw how the heatmap demonstrated a strong association between smoking and heart attack risk. Current and former smokers face significantly higher risks, especially when they do not consume alcohol, highlighting the severe health impacts of smoking. Interestingly, moderate alcohol consumption appears to reduce risk, aligning with prior research suggesting cardioprotective effects through increased "good" cholesterol. However, the dataset lacks details on alcohol consumption levels, limiting our ability to explore these effects further. This analysis emphasizes the need for public health campaigns targeting smoking cessation and responsible drinking habits.

Adding to this, the coefficient plot shows that stroke, diabetes, and kidney disease significantly elevate heart attack risks, with stroke being the most impactful. This highlights the cardiovascular system's vulnerability following a stroke and the systemic deterioration associated with chronic conditions like diabetes and kidney disease. Notably, gestational diabetes appears to reduce risk, potentially reflecting enhanced health monitoring during pregnancy.

These findings underscore the need for targeted interventions to address disparities in heart attack risks. Policies should prioritize rural healthcare access, gender-specific health education, and smoking cessation programs. Furthermore, the protective effect of moderate alcohol consumption requires more nuanced public health messaging. Addressing chronic conditions like diabetes and kidney disease with proactive medical care can significantly reduce cardiovascular risks.

While the dataset provides valuable insights, it has limitations. It does not include detailed information on alcohol consumption levels, physical activity, or socioeconomic factors, which could influence heart attack risks. The dataset is also constrained to one year (2018), limiting its ability to capture temporal trends or long-term impacts. Future studies should incorporate longitudinal data, granular lifestyle metrics, and socioeconomic variables to provide a more comprehensive understanding of heart attack risk factors.

By drawing connections between the graphs and broader societal and medical challenges, this study highlights actionable steps to mitigate heart attack risks and improve public health outcomes.

Conclusion

The steadily increasing rate of heart attacks in the US raises concerns about healthcare accessibility and effectiveness. This report explored some possible factors that can increase heart attack risk, identifying patient characteristics that may contribute to a higher likelihood of experiencing a heart attack, such as age, gender, lifestyle habits, and medical history. Our analysis shows that clinical conditions such as kidney disease, diabetes, and stroke significantly elevate the risk of heart attacks, with stroke showing the highest odds ratio among these factors. Interestingly, pregnancy-related diabetes was associated with reduced odds, potentially due to better monitoring and care during pregnancy, highlighting the role of healthcare intervention. These findings emphasize the need for targeted approaches to manage chronic conditions in individuals with a higher predisposition to heart disease. To build on the findings from this study, further research needs to be done to better understand why rural locations have higher rates of heart attack and what actions can be taken to decrease this. Increasing access to healthcare in these areas and encouraging health practices could work to reduce the likelihood. Further exploration of why men have such high rates of heart attack is critical for medical professionals to best support their patients. The heatmap also revealed that a patient's alcohol consumption may affect their chances of a heart attack, which can be studied more with data on specific consumption levels. This can gauge if certain levels of drinking reduce heart attack occurrences.

Overall, the study highlights the importance of addressing individual risk factors and systemic barriers in healthcare to reduce heart attack rates. By prioritizing further research into these areas and implementing targeted interventions, heart attack risks can be minimized in the US.